Diabetic Retinopathy Digital Twin — Architecture Diagram

CRP Digital Twin Challenge 2026 — Happiest Health

1. Data Ingestion Layer
Multi-Source Clinical Inputs

- Portable Fundus Images (JPG/PNG)
- Patient EHR (HbA1c, BP, eGFR, Meds)
- Continuous Glucose Monitor (CGM)
- Wearable Vitals (Heart Rate, Steps)

2. Vision AI & Explainability
Deep CNN & Saliency Maps

- Circular ROI Crop & Ben Graham Norm
- CLAHE Contrast Enhancement
- Pretrained EfficientNet-B3 Classifier
- 5-Class ICDR Severity Scoring (0-4)
- GradCAM Heatmap Overlay Generation
- Lesion Saliency Region Detection

3. Digital Twin Engine
Physiological & Risk Modeling

- Evidence-Based Risk Score (0-100)
- 8 Weighted Clinical Biomarkers
- 5-State Markov Stochastic Transition
- 1,000 Monte Carlo 5-Year Trajectories
- Risk-Adaptive Screening Recommender
- Personalized Lifestyle Interventions

4. Clinical UI & Output
Interactive Decision Support

- Real-Time Streamlit Web App
- Original vs GradCAM Comparison
- Plotly 0-100 Multi-Zone Risk Gauge
- 8-Axis Clinical Risk Radar Chart
- 7-Day CGM Circadian Glucose Plot
- 5-Year Progression Area Forecast